

| Tool Version : Vivado v.2025.2 (win64) Build 6299465 Fri Nov 14
19:35:11 GMT 2025
| Date : Wed Jul 1 22:37:59 2026
| Host : AYODELE running 64-bit major release (build 9200)
| Command : report_utilization -file
Matrix_Accelerator_utilization_synth.rpt -pb
Matrix_Accelerator_utilization_synth.pb
| Design : Matrix_Accelerator
| Device : xc7k70tfbv676-1
| Speed File : -1
Design State : Synthesized

Utilization Design Information

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1. Slice Logic
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Site Type	Used	Fixed	Prohibited	Available
Util%				
Slice LUTs*	89	0	0	41000
0.22				
LUT as Logic	89	0	0	41000
0.22				
LUT as Memory	0	0	0	13400
0.00				
Slice Registers	226	0	0	82000
0.28				
Register as Flip Flop	226	0	0	82000
0.28				
Register as Latch	0	0	0	82000
0.00				
F7 Muxes	0	0	0	20500
0.00				
F8 Muxes	0	0	0	10250
0.00				
Unique Control Sets	11		0	10250
0.11				

* Warning! The Final LUT count, after physical optimizations and full implementation, is typically lower. Run opt_design after synthesis, if not already completed, for a more realistic count.

Warning! LUT value is adjusted to account for LUT combining.

Warning! For any ECO changes, please run place_design if there are unplaced instances

** Note: Available Control Sets calculated as Slice * 1, Review the Control Sets Report for more information regarding control sets.

1.1 Summary of Registers by Type

Total	Clock Enable	Synchronous	Asynchronous
0	—	—	—
0	—	—	Set
0	—	—	Reset
0	—	Set	—
0	—	Reset	—
0	Yes	—	—
0	Yes	—	Set
0	Yes	—	Reset
0	Yes	Set	—
226	Yes	Reset	—

2. Memory

Site Type	Used	Fixed	Prohibited	Available	Util%
Block RAM Tile	0	0	0	135	0.00
RAMB36/FIFO*	0	0	0	135	0.00
RAMB18	0	0	0	270	0.00

* Note: Each Block RAM Tile only has one FIFO logic available and therefore can accommodate only one FIFO36E1 or one FIFO18E1. However, if a FIFO18E1 occupies a Block RAM Tile, that tile can still accommodate a RAMB18E1

3. DSP

Site Type	Used	Fixed	Prohibited	Available	Util%
DSPs	4	0	0	240	1.67
DSP48E1 only	4				

4. IO and GT Specific

Site Type	Used	Fixed	Prohibited	Available	Util%
Bonded IOB	154	0	0	300	51.33
Bonded IPADs	0	0	0	26	0.00
Bonded OPADs	0	0	0	16	0.00
PHY_CONTROL	0	0	0	6	0.00
PHASER_REF	0	0	0	6	0.00
OUT_FIFO	0	0	0	24	0.00
IN_FIFO	0	0	0	24	0.00
IDELAYCTRL	0	0	0	6	0.00
IBUFDS	0	0	0	288	0.00
GTXE2_COMMON	0	0	0		

6. Specific Feature

Site Type	Used	Fixed	Prohibited	Available	Util%
BSCANE2	0	0	0	4	0.00
CAPTUREE2	0	0	0	1	0.00
DNA_PORT	0	0	0	1	0.00
EFUSE_USR	0	0	0	1	0.00
FRAME_ECCE2	0	0	0	1	0.00
ICAPE2	0	0	0	2	0.00
PCIE_2_1	0	0	0	1	0.00
STARTUPE2	0	0	0	1	0.00
XADC	0	0	0	1	0.00

7. Primitives

Ref Name	Used	Functional Category
FDRE	226	Flop & Latch
OBUF	129	IO
LUT6	36	LUT
LUT5	33	LUT
IBUF	25	IO
LUT3	11	LUT
CARRY4	8	CarryLogic
LUT4	7	LUT
LUT2	6	LUT
DSP48E1	4	Block Arithmetic
BUFG	1	Clock

8. Black Boxes

+-----+	+-----+
Ref Name	Used
+-----+	+-----+

9. Instantiated Netlists

+-----+	+-----+
Ref Name	Used
+-----+	+-----+